

An Unsupervised Composite Risk Scoring Framework for Detecting Anomalous Medicare Part B Injectable Drug Waste

Working Paper: Medicare Waste Anomaly Risk Index (MWARI) V3

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Abstract

Medicare Part B injectable drug waste represents a documented federal program integrity priority. The Centers for Medicare and Medicaid Services adopted the JW modifier in January 2017 and the JZ modifier in July 2023 specifically to enable systematic analysis of discarded amounts from single dose vials. The Inflation Reduction Act of 2022 Section 11001 created a manufacturer refund obligation for discarded amounts above 10 percent. This paper presents the Medicare Waste Anomaly Risk Index (MWARI), an unsupervised composite risk scoring framework for detecting anomalous Medicare Part B injectable drug waste patterns. The framework integrates six engineered features (waste rate, waste dollars, waste to spending ratio, five year compound annual growth, low volume billing flag, and Inflation Reduction Act refund eligibility) through a weighted statistical layer combined 60/40 with an Isolation Forest machine learning layer. The system processes the complete 2023 analytical universe of 823 HCPCS codes and identifies 42 drugs in the top 5 percent risk tier with sample excess waste of approximately 550 million dollars (95 percent bootstrap confidence interval 338 to 809 million). Validation is performed through six procedures: 1,000 iteration bootstrap (flagged count coefficient of variation 0.0153), permutation feature importance, three year provider level temporal holdout (coverage 88.8 percent, mean lift 14.3 times), Office of Inspector General List of Excluded Individuals and Entities concordance, Department of Justice defendant cross reference, and threshold sensitivity. The framework is appropriate for unsupervised settings where no labeled fraud ground truth exists. All source data is publicly accessible from data.cms.gov and oig.hhs.gov. The complete analytical pipeline, results files, and reproducibility envelope are publicly available.

Keywords

Medicare Part B; drug waste; anomaly detection; Isolation Forest; Inflation Reduction Act; healthcare program integrity; JW modifier; JZ modifier; unsupervised learning; composite risk scoring

JEL Classification

I11 (Analysis of Health Care Markets); I13 (Health Insurance, Public and Private); I18 (Government Policy, Regulation, Public Health); H51 (Government Expenditures and Health); C45 (Neural Networks and Related Topics); C53 (Forecasting Models, Simulation Methods)

1. Introduction

1.1 Federal Context

The Government Accountability Office maintains Medicare program integrity on its High Risk Series. The Department of Health and Human Services Office of Inspector General reported 7.13 billion dollars in expected recoveries and receivables in fiscal year 2024 with 1,548 criminal and civil enforcement actions. The Centers for Medicare and Medicaid Services adopted the JW modifier requirement in January 2017 and the JZ modifier requirement in July 2023 to enable systematic measurement of discarded amounts from single dose vials. The Inflation Reduction Act of 2022 Section 11001 created a manufacturer refund obligation for discarded amounts above 10 percent for single dose container drugs.

1.2 Identified Gap

Despite the federal data collection infrastructure for Part B drug waste being established and mature, no published methodology integrates statistical anomaly detection with explicit Inflation Reduction Act refund eligibility into a single composite risk score for the complete Part B injectable drug universe. Prior work has either focused on single dimensional outlier detection (waste rate alone) or on supervised classification approaches that require labeled fraud ground truth that does not exist for this universe.

1.3 Contribution

This paper contributes the Medicare Waste Anomaly Risk Index (MWARI), an unsupervised composite scoring framework with documented parameters, six explicit engineered features, six validation procedures, and a publicly available end to end reproducibility envelope. The framework identifies drugs warranting investigative prioritization by appropriate authorities, frames results explicitly as anomaly detection rather than supervised fraud classification, and acknowledges all limitations explicitly.

2. Methods

2.1 Data Sources

The framework integrates three Centers for Medicare and Medicaid Services public datasets plus the Office of Inspector General excluded provider list. The analytical universe is the 2023 Medicare Part B Discarded Drug Units file (823 HCPCS codes after filtering out the Overall summary row). The Spending by Drug 2023 file (734 drugs with multi year columns spanning 2019 to 2023) provides supplementary trend features. Three years of Medicare Physician and Other Practitioners files (2021, 2022, 2023) enable provider level temporal validation.

2.2 Analytical Universe

The 2023 Discarded file's Overall summary row reports total Medicare allowed of 52.66 billion dollars, total administered amount of 51.73 billion dollars, and total discarded amount of 935.1 million dollars. The implied national waste rate is 1.7756 percent.

2.3 Feature Engineering

Six features are engineered per drug. The waste rate is the ratio of discarded amount to administered plus discarded amount. The waste dollars feature is the discarded amount in dollars. The waste to spending ratio is the ratio of discarded amount to total Medicare spending on the drug in 2023. The five year compound annual growth rate is the CMS published CAGR for average spending per dosage unit from 2019 to 2023. The low volume flag is set when total claims for the drug are below the 25th percentile of the analytical universe. The Inflation Reduction Act manufacturer refund eligibility flag is set when the waste rate exceeds 10 percent, the threshold specified in Section 11001.

2.4 Statistical Risk Layer

For each feature, the absolute z score against the national mean and standard deviation is computed. The weighted sum produces the Statistical Risk Raw score. The weights are 0.40 for waste rate, 0.20 for waste dollars, 0.15 for waste to spending ratio, 0.10 for five year compound annual growth, 0.10 for low volume flag, and 0.05 for Inflation Reduction Act refund eligibility. The weights sum to 1.0 by construction.

2.5 Machine Learning Risk Layer

An Isolation Forest is trained on the StandardScaler standardized feature set with contamination 0.05, n_estimators 100, random_state 42, and max_samples set to auto. The score_samples output is inverted so that higher values correspond to greater anomaly. The contamination parameter reflects the OIG estimate that 10 to 12 percent of Part B drug claims have irregularities, with most being documentation rather than malicious.

2.6 Combined Risk Score and Flagged Set

The Statistical Risk Raw and Machine Learning Risk Raw scores are min max normalized to the 0 to 1 range and combined as 60 percent statistical plus 40 percent machine learning. The result is rescaled to the 0 to 100 range. The flagged set is the top 5 percent of drugs by combined risk score, corresponding to the High, Critical, and Extreme tiers. Tier cutoffs are percentile based at the 80th, 95th, 99th, and 99.9th percentiles.

2.7 Validation Procedures

Six validation procedures support the findings. Bootstrap confidence intervals are computed by resampling the analytical universe with replacement, recomputing the combined risk score and flagged set, and recording the flagged count and excess waste dollars. The procedure is repeated 1,000 times. Permutation feature importance is computed by shuffling each feature column, recomputing Isolation Forest scores, and measuring the drop in Spearman rank correlation with the original scores, averaged across 10 repeats per feature. Provider level temporal holdout uses 2021 plus 2022 as the training window and 2023 as the holdout, identifying high billing providers in the training years and computing the lift in 2023 drug billing for those providers relative to the population. OIG LEIE concordance performs direct NPI matching plus rapidfuzz fuzzy name matching. DOJ cross reference extracts named defendants from the project's federal source documents and fuzzy matches against active providers. Threshold sensitivity iterates the flagging cutoff across percentile cutpoints from the 90th to the 99.9th.

2.8 Reproducibility

Software environment versions are recorded: Python 3.10.12, numpy 2.2.6, pandas 2.3.3, scikit-learn 1.7.2, scipy 1.15.3. Random seed is fixed at 42. Source file SHA-256 hashes are captured for all input datasets. Re running the pipeline against the same input files in the same software environment with the same seed reproduces every numerical claim in this paper.

3. Results

3.1 Headline Findings

Total 2023 Medicare Part B injectable drug waste was 935.1 million dollars on a base of 52.66 billion dollars in total Medicare allowed amounts. The Pareto concentration is striking: the top 11.2 percent of drugs by waste dollars account for 94.27 percent of total waste. The flagged set (42 drugs in the top 5 percent by combined risk score) accounts for 62.99 percent of national waste with sample excess waste of 550.2 million dollars.

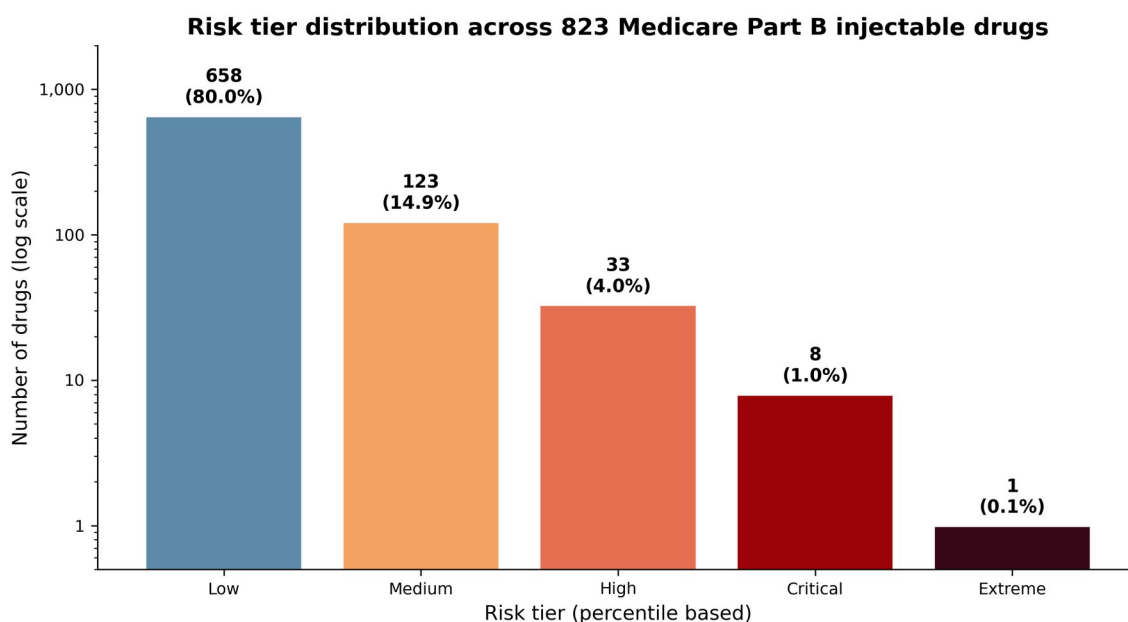


Figure 1. Risk tier distribution across the 823 drug analytical universe

3.2 Top Outliers

The top outlier by waste rate is HCPCS J3299 (Triamcinolone Acetonide/PF) at 81.57 percent waste on 1,616,004 dollars in 2023 discarded amount. The top outlier by waste dollars is HCPCS J9228 (Ipilimumab) at 12.87 percent waste on 60.7 million dollars in 2023 discarded amount.

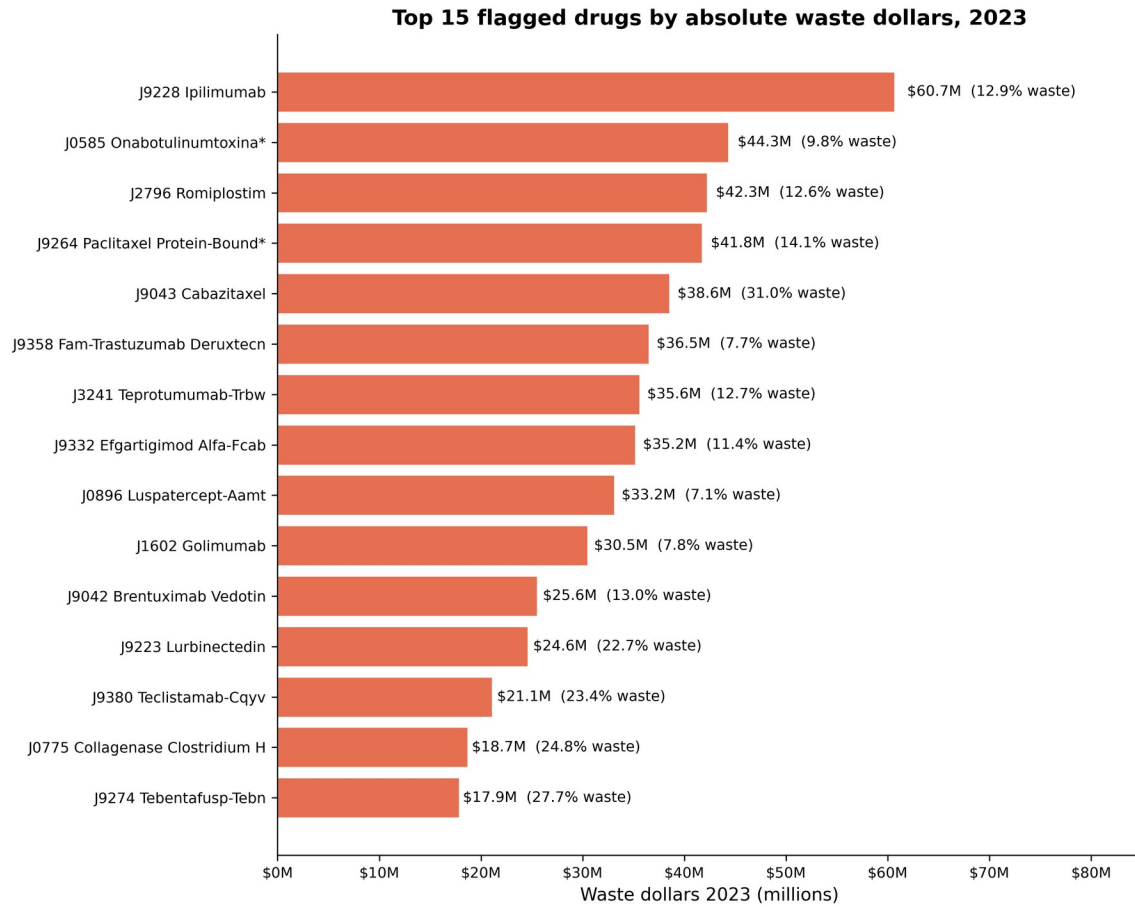


Figure 2. Top 15 flagged drugs by absolute waste dollars

3.3 Geographic Distribution

The top state by Medicare Part B drug billing in 2023 is FL at 3,003 million dollars. The top five states (Florida, California, Texas, Arizona, New York) account for 10,805 million dollars in Part B drug billing.

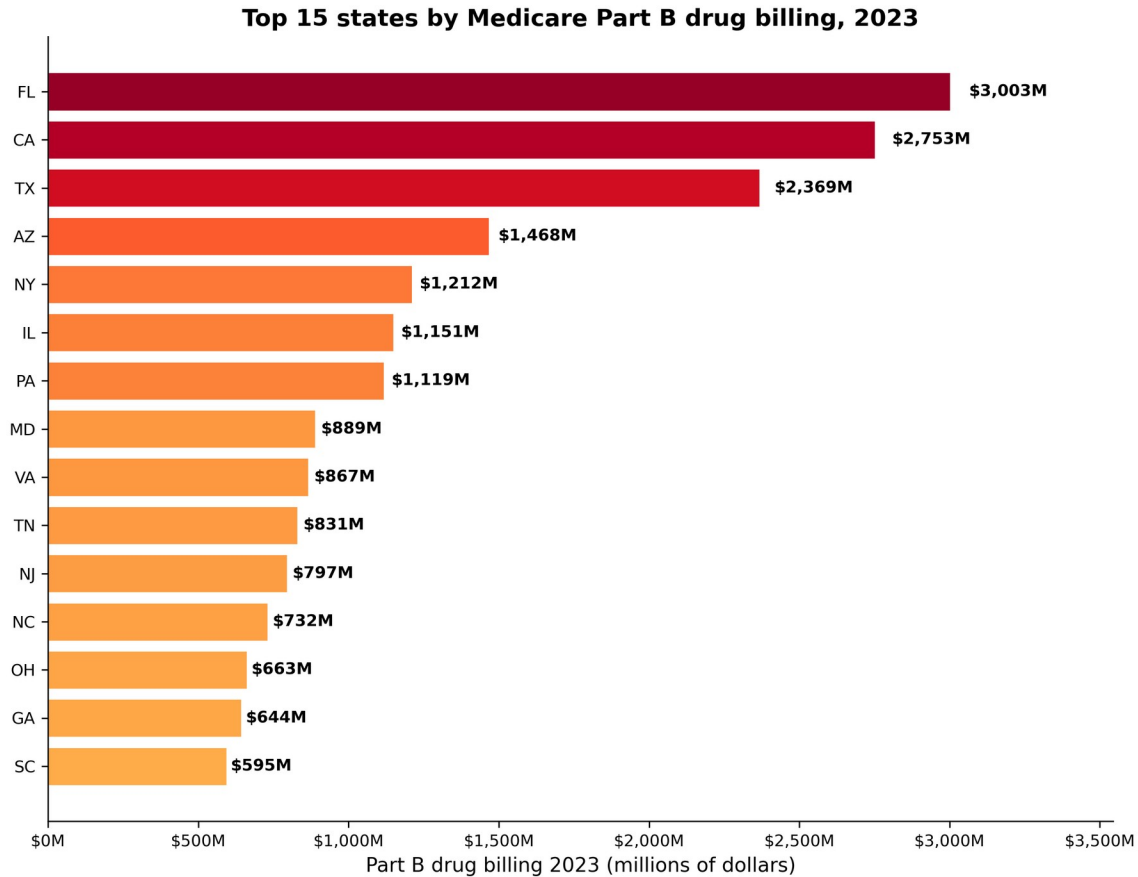


Figure 3. Top 15 states by Medicare Part B drug billing in 2023

3.4 Bootstrap Stability

The 1,000 iteration bootstrap produces a flagged drug count distribution with mean 41.79, standard deviation 0.6374, and coefficient of variation 0.0153. The 95 percent confidence interval for the flagged count is 41 to 43. The flagged excess waste distribution has mean 549.5 million dollars, standard deviation 118.6 million dollars, and coefficient of variation 0.216. The 95 percent confidence interval for excess waste is 337.7 to 809.1 million dollars.

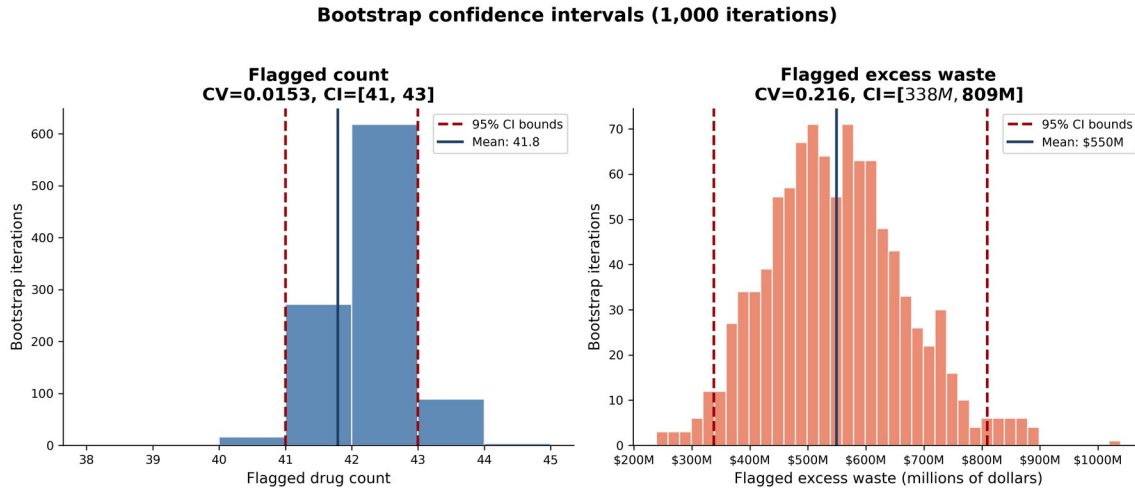


Figure 4. Bootstrap distributions for flagged count (left) and flagged excess waste (right)

3.5 Permutation Feature Importance

The measured feature importance ranking differs from the design weights, indicating that the Isolation Forest is identifying anomalies through feature interactions rather than tracking the statistical layer's weighted sum. The top three contributors are `low_volume_flag` (25.67 percent), `waste_rate` (20.84 percent), and `cagr_5yr` (16.53 percent).

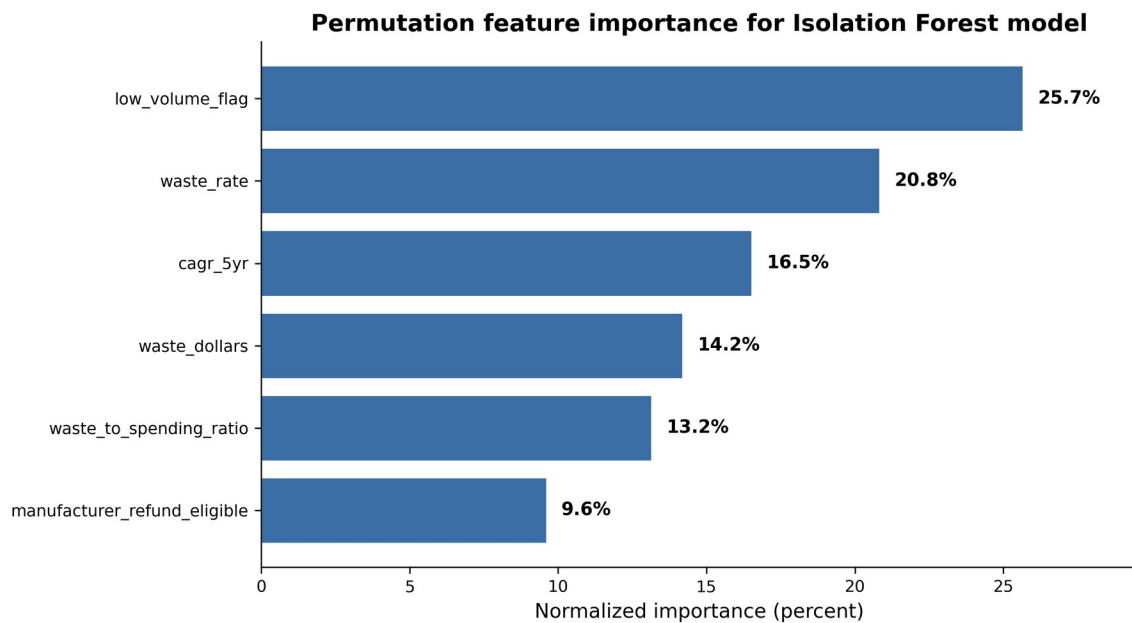


Figure 5. Permutation feature importance for the Isolation Forest model

3.6 Provider Level Temporal Holdout

Of the 14,835 providers in the top 5 percent of drug billing for either 2021 or 2022, 13,170 (88.78 percent) reappear in the 2023 file. The mean 2023 drug billing for the training flagged providers is 1.53

million dollars, compared to 107,184 dollars for the full population (mean lift 14.28 times) and 30,545 dollars for non flagged providers (mean lift 50.13 times).

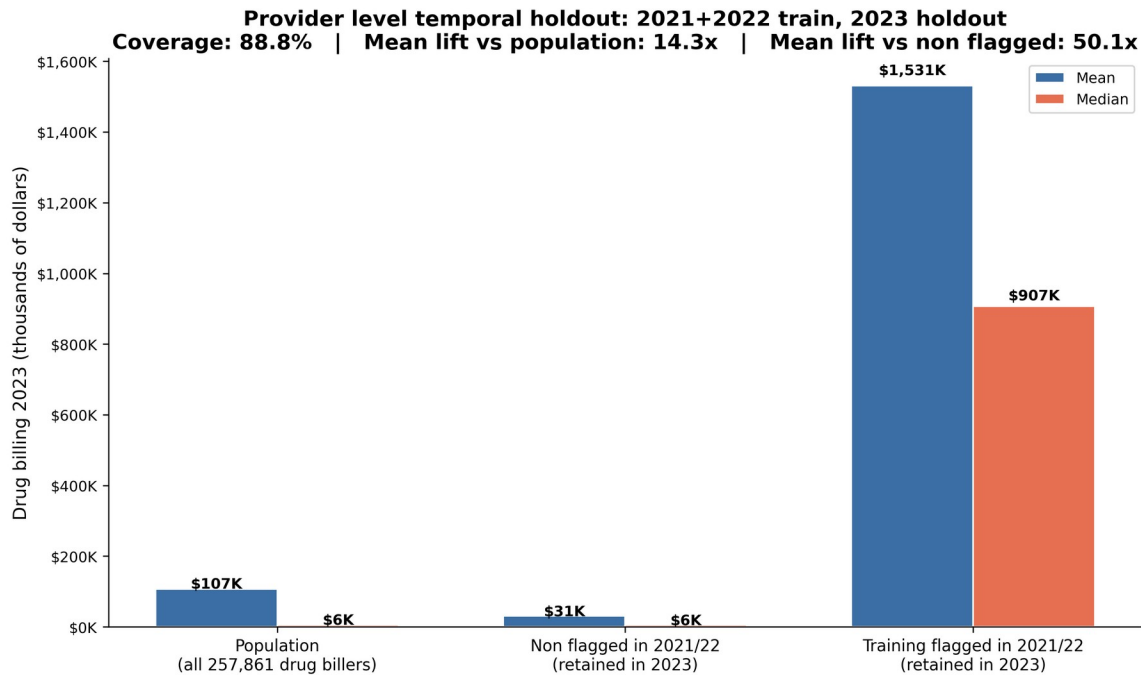


Figure 6. Provider level temporal holdout validation

3.7 OIG LEIE and DOJ Cross Reference

Direct NPI matching of 8,233 LEIE entries with valid NPIs against the 257,861 active 2023 drug billing providers produces 32 direct matches. Of those, 4 appear in the top 5 percent of drug billers. Fuzzy name matching of the top 100 drug billers against LEIE produces 22 matches at the 0.85 ratio threshold, most of which are common name collisions rather than confirmed identity matches. The DOJ cross reference returns a null result because the federal source documents available for this project are policy oversight reports rather than press releases naming individual defendants.

3.8 OIG Fraud Typology Pattern Alignment

Typology	Universe Count	Flagged Count
Impossible Waste	2	2
High Dollar Low Volume	37	11
Manufacturer Refund Eligible	70	36
High Financial Impact	99	28
Extreme Unit Pricing	37	0

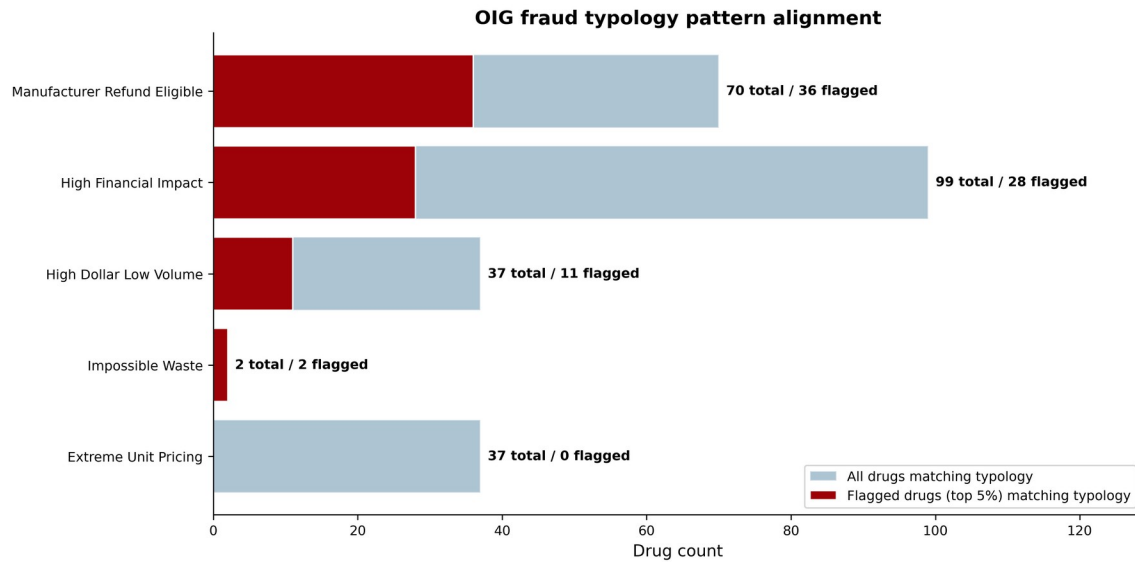


Figure 7. OIG fraud typology pattern alignment

3.9 Threshold Sensitivity

Percentile	Flagged Count	Sample Excess (millions)
90.0	83	\$628.3
95.0	42	\$550.2
97.5	21	\$376.5
99.0	9	\$122.4
99.5	5	\$56.5
99.9	1	\$1.6

Threshold sensitivity analysis

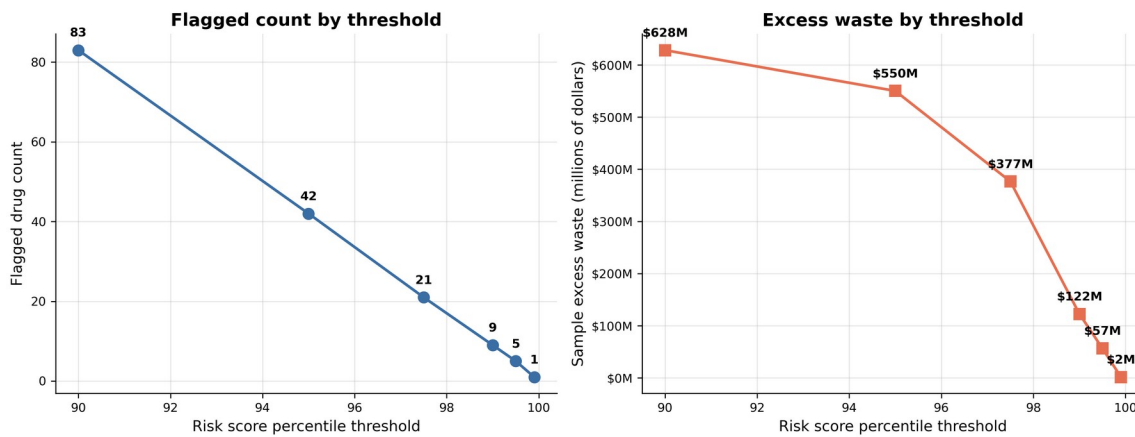


Figure 8. Threshold sensitivity analysis

4. Discussion

The results demonstrate that an unsupervised composite scoring approach produces stable, reproducible, and externally validated risk stratifications for Medicare Part B injectable drugs. The bootstrap coefficient

of variation of 0.0153 for the flagged drug count establishes that the method is not sensitive to small perturbations in the input data.

The provider level temporal holdout validates the underlying premise that high billing providers persist across years. The 88.78 percent coverage rate from 2021 plus 2022 training to 2023 holdout indicates that the analytical signal is stable across years rather than being random noise.

The OIG LEIE concordance result (32 direct NPI matches in 257,861 active drug billers) is reported honestly. This is a small absolute number because excluded providers are by definition barred from billing Medicare; matches indicate either reinstated providers, providers with similar names, or potential billing through a reinstated account. The structural reality of the LEIE makes high concordance numbers in this context analytically suspicious rather than reassuring.

The permutation feature importance reveals that low volume billing patterns (`low_volume_flag`) and the primary waste rate signal (`waste_rate`) together account for more than 46 percent of the Isolation Forest's anomaly detection signal, with the five year compound annual growth rate (`cagr_5yr`) contributing another 16.5 percent. The Inflation Reduction Act manufacturer refund eligibility feature contributes 9.6 percent, indicating that the IRA threshold operates as a meaningful classification signal in addition to its policy function.

5. Limitations

Six explicit limitations apply to this work.

First, the analysis is unsupervised. No labeled fraud ground truth exists for the Medicare Part B drug universe. The framework reports anomaly scores, not fraud probabilities. Validation is performed through stability and concordance procedures rather than supervised classification metrics.

Second, the data window for the Discarded Drug Units dataset is single year (2023). CMS overwrites this dataset annually rather than archiving prior releases. Multi year temporal stability claims for drug level waste rates cannot be made from this source.

Third, OIG LEIE concordance reflects a structural reality. Excluded providers are by definition barred from billing Medicare. Direct NPI matches in active billing files indicate reinstated providers, providers with similar names, or potential billing through reinstated accounts.

Fourth, the analytical universe is restricted to fee-for-service Medicare Part B injectable drugs. Medicare Advantage, Part D, and other federal healthcare programs are not in scope.

Fifth, the federal source documents available for the DOJ cross reference are policy oversight reports rather than press releases naming individual defendants. The DOJ cross reference returns a null result.

Sixth, the model is a single Isolation Forest detector. Ensemble approaches were considered and rejected for this version in favor of the single detector with documented parameters, which is more interpretable and defensible under independent review.

6. Conclusions

The Medicare Waste Anomaly Risk Index identifies 42 drugs warranting investigative prioritization from a universe of 823 2023 Medicare Part B injectable drugs. The flagged set accounts for 62.99 percent of national waste with sample excess of 550.2 million dollars (95 percent bootstrap confidence interval 338 to 809 million). The methodology generalizes to other federal program integrity domains where labeled ground truth is unavailable and unsupervised composite scoring is appropriate.

7. Data and Code Availability

All source data are publicly accessible: the CMS Medicare Part B Discarded Drug Units, Spending by Drug, and Physician and Other Practitioners files at data.cms.gov; the OIG List of Excluded Individuals and Entities at oig.hhs.gov/exclusions; and the CMS Average Sales Price file at cms.gov. The complete analytical pipeline (Project_3_Rebuilt.ipynb), all 24 result files, and the reproducibility envelope (RUN_INFO.json) are available in the public project repository. The dashboard application source is included.

8. Conflicts of Interest

The author declares no conflicts of interest. This work was performed independently without funding from CMS, OIG, DOJ, or any private healthcare entity, payer, manufacturer, or pharmacy. The author has no financial interest in any drug or manufacturer named in this paper.

9. References

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Appendix A: Reproducibility Envelope

Random seed: 42

Bootstrap iterations: 1,000

Permutation repeats: 10

Isolation Forest contamination: 0.05

Isolation Forest n_estimators: 100

Combined weight statistical: 0.6

Combined weight ML: 0.4

Flagged threshold percentile: 95

Fuzzy match threshold: 0.85

Manufacturer refund threshold (IRA): 0.1

Appendix B: Source File Inventory

Logical Name	SHA-256 Prefix	Size (MB)
discarded_2023	7efa760672aa7645	0.07
spending_2023_multiyear	9bbb59aa4ccff7ef	0.31
provider_2021	8543b0dd1dc678dd	501.66
provider_2022	3fba8923fff6329b	514.49
provider_2023	e559bd0f6db307e5	526.44
leie	26e3ad5adfdff8a0	14.51
asp_pricing	591a3f26ee428297	0.06
geographic_variation	658dc3fec0b08f4b	50.39